Ps 02

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1 Problem 18 DONE

1.1 Task

Use the Master Theorem to provide Θ bounds for the following recursions (assume $n_0 = 1$ for all of them).

(a) $T(n) = 2T\left(\frac{n}{4}\right) + \Theta(1)$,

(b) $T(n) = 2T\left(\frac{n}{4}\right) + \Theta(\sqrt{n})$,

(c) $T(n) = 2T\left(\frac{n}{4}\right) + \Theta(\sqrt{n}\lg^2 n)$,

(d) $T(n) = 2T\left(\frac{n}{4}\right) + \Theta(n)$,

(e) $T(n) = 2T\left(\frac{n}{4}\right) + \Theta(n^2)$.

1.2 Solution

1.2.1 (a)

$$\begin{aligned} a &= 2 \quad \text{(number of subproblems)} \\ b &= 4 \quad \text{(factor by which input shrinks)} \\ d &= 0 \quad \text{(since the non recursive block is } \Theta(n^0) = \Theta(1)) \end{aligned}$$

$$\begin{aligned} \text{since } d &< \log_b a \\ 0 &< \log_4 2 \\ 0 &< \frac{1}{2} \end{aligned}$$

It is a Master Theorem Case 1, such that

$$T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_4 2}) = \Theta(\sqrt{n})$$

1.2.2 (b)

$$\begin{aligned} a &= 2 \quad \text{(number of subproblems)} \\ b &= 4 \quad \text{(factor by which input shrinks)} \\ d &= \frac{1}{2} \quad \text{(since the non recursive block is } \Theta(\sqrt{n}) = \Theta(n^{1/2})) \end{aligned}$$

$$\begin{aligned} \text{since } d &= \log_b a \\ \frac{1}{2} &= \log_4 2 \\ \frac{1}{2} &= \frac{1}{2} \end{aligned}$$

It is a Master Theorem Case 2, such that

$$T(n) = \Theta(n^{\log_b a} \lg^{k+1} n) = \Theta(n^{\log_4 2} \lg n) = \Theta(\sqrt{n} \lg n)$$

1.2.3 (c)

$$\begin{aligned} a &= 2 \quad \text{(number of subproblems)} \\ b &= 4 \quad \text{(factor by which input shrinks)} \end{aligned}$$

Notice that $f(n) = \Theta(\sqrt{n} \lg^2 n) = \Theta(n^{1/2} \lg^2 n)$. The polynomial part equals $n^{\log_b a}$. The extra $\lg^2 n$ factor matches Case 2 pattern.

$$\begin{aligned} f(n) &= \Theta(n^{\log_b a} \lg^k n) \quad \text{with } k = 2 \\ \text{Hence, } T(n) &= \Theta(n^{\log_b a} \lg^{k+1} n) \\ &= \Theta(n^{1/2} \lg^{2+1} n) \\ &= \Theta(\sqrt{n} \lg^3 n) \end{aligned}$$

1.2.4 (d)

$$\begin{aligned} a &= 2 \quad \text{(number of subproblems)} \\ b &= 4 \quad \text{(factor by which input shrinks)} \\ d &= 1 \quad \text{(since the non recursive block is } \Theta(n) = \Theta(n^1)\text{)} \end{aligned}$$

$$\begin{aligned} \text{since } d &> \log_b a \\ 1 &> \log_4 2 \\ 1 &> \frac{1}{2} \end{aligned}$$

It is a Master Theorem Case 3, such that

$$T(n) = \Theta(f(n)) = \Theta(n)$$

1.2.5 (e)

$$\begin{aligned} a &= 2 \quad \text{(number of subproblems)} \\ b &= 4 \quad \text{(factor by which input shrinks)} \\ d &= 2 \quad \text{(since the non recursive block is } \Theta(n^2)\text{)} \end{aligned}$$

$$\begin{aligned} \text{since } d &> \log_b a \\ 2 &> \log_4 2 \\ 2 &> \frac{1}{2} \end{aligned}$$

It is a Master Theorem Case 3, such that

$$T(n) = \Theta(f(n)) = \Theta(n^2)$$

2 Problem 19 DONE

2.1 Task

Professor Caesar wants to develop a matrix-multiplication algorithm that is asymptotically faster than Strassen's algorithm. His algorithm will use the divide-and-conquer method, dividing each matrix into submatrices of size $\frac{n}{4} \times \frac{n}{4}$, and the divide and combine steps together will take $\Theta(n^2)$ time. Suppose that the professor's algorithm creates a recursive subproblems of that size. What is the largest integer value of a for which his algorithm could possibly run asymptotically faster than Strassen's? How does a change if the subproblems have size $\frac{n}{3} \times \frac{n}{3}$ or $\frac{n}{5} \times \frac{n}{5}$?

2.2 Solution

Strassen algorithm is:

$$T(n) = 7\,T(n/2) + \Theta(n^2) \implies T(n) = \Theta(n^{\lg 7}) \approx \Theta(n^{2.807})$$

- Subproblem size $n/4 \times n/4$

  Professor Caesar's algorithm is:

  $$T(n) = aT(n/4) + \Theta(n^2) \implies T(n) = \Theta(n^{\log_4 a})$$

  Assume Caesar's algorithm outperforms Strassen's. Hence,

  $$\begin{aligned} n^{\log_4 a} &< n^{\lg 7} \\ \implies \log_4 a &< \log_2 7 \\ \implies a &< 4^{\log_2 7} \\ &< (2^2)^{\log_2 7} \\ &< 2^{2\log_2 7} \\ &< 7^2 = 49 \end{aligned}$$

  Hence, the largest integer value of a is 48.

- Subproblem size $n/3 \times n/3$

  $$T(n) = aT(n/3) + \Theta(n^2) \implies T(n) = \Theta(n^{\log_3 a})$$

$$
\begin{aligned}
\log_3 a &< \log_2 7 \\
\implies a &< 3^{\log_2 7} \\
&< 3^{2.807\ldots} \\
&< 21.85\ldots
\end{aligned}
$$

Hence, the largest integer value of a is 21.

- Subproblem size $n/5 \times n/5$

$$T(n) = aT(n/5) + \Theta(n^2) \implies T(n) = \Theta(n^{\log_5 a})$$

$$
\begin{aligned}
\log_5 a &< \log_2 7 \\
\implies a &< 5^{\log_2 7} \\
&< 5^{2.807\ldots} \\
&< 70.18\ldots
\end{aligned}
$$

Hence, the largest integer value of a is 70.

3 Problem 20 DONE

3.1 Task

Consider the function $f(n) = \lg n$. Argue that although $f\left(\frac{n}{2}\right) < f(n)$, the regularity condition $af\left(\frac{n}{b}\right) \leq cf(n)$ with $a = 1$ and $b = 2$ does not hold for any constant $c < 1$. Argue further that for any $\varepsilon > 0$, the condition in Case 3 of the Master theorem that $f(n) = \Omega(n^{\log_b a+\varepsilon})$ does not hold.

3.2 Solution

- Part 1

  We need to check whether $af(\frac{n}{b}) \leq cf(n)$ holds for $a = 1, b = 2$ and $f(n) = \lg n$ for any constant $c < 1$.

$$
\begin{aligned}
af(\frac{n}{b}) &\leq cf(n) \\
1 \cdot f(\frac{n}{2}) &\leq cf(n) \\
\lg \frac{n}{2} &\leq c \lg n \\
\lg n - \lg 2 &\leq c \lg n \\
\lg n - 1 &\leq c \lg n \\
(1-c) \lg n &\leq 1 \\
\lg n &\leq \frac{1}{1-c}
\end{aligned}
$$

But $\lg n$ grows without bound as $n \to \infty$. For a fixed $c < 1$, this could not hold for all large n.

Hence, the regularity condition fails $\forall c < 1$.

- Part 2

  Case 3 of Master Theorem requires $\exists \epsilon > 0 : f(n) = \Omega(n^{\log_b a + \epsilon})$.

  With $a = 1, b = 2$, $\log_b a = \log_2 1 = 0$.

  So the Case 3 would require:

  $$f(n) = \Omega(n^{\epsilon})$$

  for some $\epsilon > 0$.

  However, logarithmic function could not grow polynomially, no matter hom small $\epsilon > 0$ is.

  $$\lim_{n \to \infty} \frac{\lg n}{n^{\epsilon}} = 0$$

  Thus,

  $$f(n) \neq \Omega(n^{\epsilon})$$

  for any $\epsilon > 0$.

4 Problem 21 DONE

4.1 Task

Show that for suitable constants $a > 0$, $b > 1$, $\varepsilon > 0$ the function $f(n) := 2^{\lceil \lg n \rceil}$ satisfies all conditions of Case 3 of the Master theorem except for the regularity condition.

4.2 Solution

Master Theorem Case 3 requires two conditions to hold for $f(n)$:

- Dominance condition: $f(n) = \Omega(n^{\log_b a+\varepsilon})$ for some $\varepsilon > 0$.
- Regularity condition: $af(n/b) \leq cf(n)$ for some constant $c < 1$ and all sufficiently large n.

Choose the constants $a = 1$, $b = 1.5$, and $\varepsilon = 0.5$. These satisfy the problem's requirements since $a > 0$, $b > 1$, and $\varepsilon > 0$.

- Dominance Condition

  Show: $f(n) = \Omega(n^{\log_b a+\varepsilon})$.

  First, note that $\log_{1.5} 1 = 0$. Thus, substituting our constants, we need to show $f(n) = \Omega(n^{0.5})$.

  We bound $f(n) = 2^{\lceil \lg n \rceil}$ to find its asymptotic growth:

$$\begin{aligned} \lg n \leq & \lceil \lg n \rceil \leq \lg n + 1 \\ 2^{\lg n} \leq & 2^{\lceil \lg n \rceil} \leq 2^{\lg n+1} \\ n \leq & 2^{\lceil \lg n \rceil} \leq 2n \end{aligned}$$

  Hence, $f(n) = \Theta(n)$.

  Since $f(n) = \Theta(n)$, it grows strictly faster than $n^{0.5}$, meaning:

$$f(n) = \Omega(n^{0.5})$$

  The dominance condition holds.

- Regularity Condition

  Show: $af(n/b) \leq cf(n)$ fails for any $c < 1$.

  Substituting our chosen constants and $f(n)$, we check if the following inequality can hold for $c < 1$:

$$1 \cdot 2^{\lceil \lg(n/1.5) \rceil} \leq c \cdot 2^{\lceil \lg n \rceil}$$

  To demonstrate that this fails, we only need to find a subsequence of n where the inequality breaks down. Consider the sequence where n is an exact power of 2, i.e., $n = 2^k$ for integers $k \geq 1$.

  For these values, $\lceil \lg n \rceil = \lceil \lg(2^k) \rceil = k$, so:

$$f(n) = 2^k$$

  Now evaluate $f(n/1.5)$ for this same subsequence:

$$f(n/1.5) = 2^{\lceil \lg(2^k/1.5) \rceil} = 2^{\lceil k - \lg 1.5 \rceil}$$

Since $\lg 1.5 \approx 0.58$, the fractional part subtracted from k is small enough that the ceiling function rounds it right back up to k. Specifically, $\lceil k - 0.58 \rceil = k$. Therefore:

$$f(n/1.5) = 2^k = f(n)$$

Substituting this back into the regularity condition yields:

$$f(n) \leq cf(n) \implies 1 \leq c$$

This directly contradicts the requirement that $c < 1$. Therefore, the regularity condition fails for all $c < 1$.

5 Problem 22 SKIP

5.1 Task

Show that $f(n) = n^2$ satisfies the polynomial growth condition but $g(n) = 2^n$ does not.

5.2 Solution

Definition: A function $h(n)$ has polynomial growth if $\exists k > 0$ such that $h(n) = O(n^k)$.

- $f(n) = n^2$:

  Let $k = 2$. By definition of Big-O, we need $c > 0$ and $n_0 > 0$ such that $n^2 \leq c \cdot n^k$ for all $n \geq n_0$.

  $$n^2 \leq 1 \cdot n^2 \quad \forall n \geq 1$$

  Setting $c = 1$ and $n_0 = 1$ satisfies the condition.

  $$\implies f(n) = O(n^2)$$

  Therefore, $f(n)$ satisfies the polynomial growth condition.

- $g(n) = 2^n$:

  Assume for contradiction that $\exists k > 0$ such that $2^n = O(n^k)$. This implies that $\limsup_{n\to\infty} \frac{2^n}{n^k} < \infty$.

  Evaluate the limit using L'Hôpital's rule. Let $m = \lceil k \rceil$:

$$\lim_{n\to\infty} \frac{2^n}{n^k} = \lim_{n\to\infty} \frac{\frac{d^m}{dn^m}(2^n)}{\frac{d^m}{dn^m}(n^k)}$$

The m th derivative of the numerator is $2^n(\ln 2)^m$. Since $m \geq k$, the m th derivative of the denominator n^k is a constant C (where $C = k!$ if $k \in \mathbb{N}$, or 0 otherwise).

$$\lim_{n\to\infty} \frac{2^n(\ln 2)^m}{C} = \infty$$

$$\implies 2^n = \omega(n^k)$$

This contradicts $2^n = O(n^k)$. Thus, $2^n \neq O(n^k)$ for any $k > 0$, and $g(n)$ does not satisfy the polynomial growth condition.

6 Problem 23 SKIP

6.1 Task

Prove that the polynomial growth condition is equivalent to the following condition on the function $f : \mathbb{R} \to \mathbb{R}$:

For all $0 < \varphi < 1$ there exist constants $N > 0$ and $c_1, c_2 > 0$ with the property that for all $n \geq N$ and all $u \in [\varphi n, n]$

$c_1 f(n) \leq f(u) \leq c_2 f(n)$.

6.2 Solution

7 Problem 24 DONE

7.1 Task

Prove that $f(n) = n \lg n$ and $f(n) = n \lg \lg n$ satisfy the polynomial growth condition.

7.2 Solution

Definition: A function $f(n)$ has polynomial growth if $\exists k > 0$ such that $f(n) = O(n^k)$.

- $f(n) = n \lg n$:

  Choose $k = 2$. We know that $\lg n < n$ for all $n \geq 1$.

$$n \lg n \leq n \cdot n = n^2 \quad \forall n \geq 1$$

By the definition of Big-O, setting $c = 1$ and $n_0 = 1$:

$$n \lg n \leq 1 \cdot n^2 \implies n \lg n = O(n^2)$$

Therefore, $n \lg n$ is polynomially bounded and satisfies the condition.

- $f(n) = n \lg \lg n$:

  Choose $k = 2$. For $n \geq 4$, the iterated logarithm is well-defined and strictly bounded by the logarithm, so $\lg \lg n < \lg n < n$.

  $$n \lg \lg n < n \cdot n = n^2 \quad \forall n \geq 4$$

  By the definition of Big-O, setting $c = 1$ and $n_0 = 4$:

  $$n \lg \lg n \leq 1 \cdot n^2 \implies n \lg \lg n = O(n^2)$$

  Therefore, $n \lg \lg n$ is polynomially bounded and satisfies the condition.

8 Problem 25 SKIP

8.1 Task

Assume that f satisfies the polynomial growth condition, and let $g(n) = \Theta(n)$. Show that $f(g(n)) = \Theta(f(n))$. Usually we write $f(\Theta(n)) = \Theta(f(n))$. Explain why this notation makes sense.

8.2 Solution

9 Problem 26 DONE

9.1 Task

Use the Akra-Bazzi-Theorem to compute Θ expressions for the following recurrences:

(a) $T(n) = T\left(\frac{n}{2}\right) + T\left(\frac{n}{3}\right) + T\left(\frac{n}{6}\right) + \Theta(n \lg n)$,

(b) $T(n) = 3T\left(\frac{n}{3}\right) + 8T\left(\frac{n}{4}\right) + \Theta\left(\frac{n^2}{\lg n}\right)$,

(c) $T(n) = \frac{2}{3}T\left(\frac{n}{3}\right) + \frac{1}{3}T\left(\frac{2n}{3}\right) + \Theta(\lg n)$,

(d) $T(n) = \dfrac{1}{3}T\left(\dfrac{n}{3}\right) + \dfrac{1}{n}$,

(e) $T(n) = 3T\left(\dfrac{n}{3}\right) + 3T\left(\dfrac{2n}{3}\right) + n^2$.

9.2 Solution

Akra-Bazzi Theorem: For $T(n) = \sum_{i=1}^{k} a_i T(b_i n) + g(n)$, find p such that $\sum_{i=1}^{k} a_i b_i^p = 1$. Then, $T(n) = \Theta\left(n^p\left(1 + \int_1^n \frac{g(u)}{u^{p+1}} du\right)\right)$.

- **(a)** $T(n) = T\left(\dfrac{n}{2}\right) + T\left(\dfrac{n}{3}\right) + T\left(\dfrac{n}{6}\right) + \Theta(n \lg n)$

  $\left(\frac{1}{2}\right)^p + \left(\frac{1}{3}\right)^p + \left(\frac{1}{6}\right)^p = 1 \implies p = 1$

  $g(n) = n \lg n$

  $\int_1^n \frac{u \lg u}{u^2} du = \int_1^n \frac{\lg u}{u} du = \Theta(\lg^2 n)$

  $T(n) = \Theta\left(n^1(1 + \lg^2 n)\right) = \Theta(n \lg^2 n)$

- **(b)** $T(n) = 3T\left(\dfrac{n}{3}\right) + 8T\left(\dfrac{n}{4}\right) + \Theta\left(\dfrac{n^2}{\lg n}\right)$

  $3\left(\frac{1}{3}\right)^p + 8\left(\frac{1}{4}\right)^p = 1$. Observe that for $p = 2$, $3(1/9) + 8(1/16) = 5/6 < 1$. Thus, $p < 2$.

  $g(n) = \frac{n^2}{\lg n}$

  $\int_1^n \frac{u^2/\lg u}{u^{p+1}} du = \int_1^n \frac{u^{1-p}}{\lg u} du = \Theta\left(\frac{n^{2-p}}{\lg n}\right)$ (since $1 - p > -1$)

  $T(n) = \Theta\left(n^p\left(1 + \frac{n^{2-p}}{\lg n}\right)\right) = \Theta\left(n^p + \frac{n^2}{\lg n}\right) = \Theta\left(\frac{n^2}{\lg n}\right)$

- **(c)** $T(n) = \dfrac{2}{3}T\left(\dfrac{n}{3}\right) + \dfrac{1}{3}T\left(\dfrac{2n}{3}\right) + \Theta(\lg n)$

  $\frac{2}{3}\left(\frac{1}{3}\right)^p + \frac{1}{3}\left(\frac{2}{3}\right)^p = 1 \implies p = 0$

  $g(n) = \lg n$

  $\int_1^n \frac{\lg u}{u^1} du = \Theta(\lg^2 n)$

  $T(n) = \Theta\left(n^0(1 + \lg^2 n)\right) = \Theta(\lg^2 n)$

- **(d)** $T(n) = \dfrac{1}{3}T\left(\dfrac{n}{3}\right) + \dfrac{1}{n}$

  $\frac{1}{3}\left(\frac{1}{3}\right)^p = 1 \implies \left(\frac{1}{3}\right)^{p+1} = 1 \implies p = -1$

  $g(n) = \frac{1}{n}$

  $\int_1^n \frac{1/u}{u^0} du = \int_1^n \frac{1}{u} du = \Theta(\lg n)$

  $T(n) = \Theta\left(n^{-1}(1 + \lg n)\right) = \Theta\left(\frac{\lg n}{n}\right)$

- **(e)** $T(n) = 3T\left(\frac{n}{3}\right) + 3T\left(\frac{2n}{3}\right) + n^2$

  $3\left(\frac{1}{3}\right)^p + 3\left(\frac{2}{3}\right)^p = 1$. Observe that for $p = 2$, $3(1/9) + 3(4/9) = 15/9 > 1$. Thus, $p > 2$.

  $g(n) = n^2$

  $\int_1^n \frac{u^2}{u^{p+1}} du = \int_1^n u^{1-p} du$. Since $p > 2 \implies 1 - p < -1$, the integral converges to a constant $\Theta(1)$ as $n \to \infty$.

  $T(n) = \Theta\left(n^p(1 + \Theta(1))\right) = \Theta(n^p)$ (where p is the solution to $3^{1-p} + 3(2/3)^p = 1$)

10 Problem 27 SKIP

10.1 Task

Use the Akra-Bazzi-Theorem for finding another proof for the continuous Master Theorem.

10.2 Solution

11 Problem 28 DONE

11.1 Task

Solving recurrences using a change of variables. Let us consider the recurrence $T(n) = 2T(\sqrt{n}) + \Theta(\lg n)$. For this recurrence the Master Theorem cannot be used directly. Therefore, define $m = \lg n$ and $S(m) = T(2^m)$. Rewrite the recurrence for T in n into a recurrence for S in m. Use the Master Theorem to solve the recurrence for S and rewrite the solution in terms of T and n.

11.2 Solution

- **Change of variables:** Let $m = \lg n \implies n = 2^m$.

  Substitute $n = 2^m$ into the original recurrence:

$$T(2^m) = 2T(\sqrt{2^m}) + \Theta(\lg(2^m))$$
$$T(2^m) = 2T(2^{m/2}) + \Theta(m)$$

- **Define new function:** Let $S(m) = T(2^m)$.

  Substitute $S(m)$ into the equation:

$$S(m) = 2S(m/2) + \Theta(m)$$

- **Apply Master Theorem to $S(m)$:**
  For $S(m) = aS(m/b) + f(m)$, we have $a = 2$, $b = 2$, and $f(m) = \Theta(m)$.
  Calculate the critical exponent:

$$m^{\log_b a} = m^{\log_2 2} = m^1 = m$$

  Compare $f(m)$ to $m^{\log_b a}$:

$$f(m) = \Theta(m) = \Theta(m^{\log_2 2})$$

  This matches Case 2 of the Master Theorem. Thus:

$$S(m) = \Theta(m \lg m)$$

- **Rewrite in terms of T and n:**
  Substitute $m = \lg n$ and $S(m) = T(n)$ back into the solution:

$$T(n) = \Theta(\lg n \cdot \lg(\lg n))$$
$$T(n) = \Theta(\lg n \lg \lg n)$$

12 Problem 29 DONE

12.1 Task

Use the method of problem 28 to solve the following recurrences:
(a) $T(n) = 2T(\sqrt{n}) + \Theta(1)$,
(b) $T(n) = 3T(\sqrt[3]{n}) + \Theta(n)$,
(c) $T(n) = 3T(\sqrt{n}) + 3T(\sqrt[3]{n}) + \Theta((\lg \lg n)^2)$.

12.2 Solution

- **Change of variables:** Let $m = \lg n \implies n = 2^m$, and define $S(m) = T(2^m)$.
- **(a)** $T(n) = 2T(\sqrt{n}) + \Theta(1)$
  Substitute $n = 2^m$:

$$T(2^m) = 2T(2^{m/2}) + \Theta(1) \implies S(m) = 2S(m/2) + \Theta(1)$$

Apply Master Theorem: $a = 2, b = 2, f(m) = \Theta(1)$. Critical exponent: $m^{\log_b a} = m^{\log_2 2} = m^1$. Since $f(m) = O(m^{1-\epsilon})$ for $\epsilon = 1$, Case 1 applies.

$$S(m) = \Theta(m)$$

Back-substitute $m = \lg n$:

$$T(n) = \Theta(\lg n)$$

- **(b)** $T(n) = 3T(\sqrt[3]{n}) + \Theta(n)$

  Substitute $n = 2^m$:

$$T(2^m) = 3T(2^{m/3}) + \Theta(2^m) \implies S(m) = 3S(m/3) + \Theta(2^m)$$

  Apply Master Theorem: $a = 3, b = 3, f(m) = \Theta(2^m)$. Critical exponent: $m^{\log_3 3} = m^1$. Since $f(m) = 2^m = \Omega(m^{1+\epsilon})$ for any $\epsilon > 0$, Case 3 applies. Regularity check: $af(m/b) \leq cf(m) \implies 3 \cdot 2^{m/3} \leq c \cdot 2^m \implies 3 \cdot 2^{-2m/3} \leq c$. This holds for $c < 1$ for sufficiently large m.

$$S(m) = \Theta(2^m)$$

  Back-substitute $m = \lg n$:

$$T(n) = \Theta(2^{\lg n}) = \Theta(n)$$

- **(c)** $T(n) = 3T(\sqrt{n}) + 3T(\sqrt[3]{n}) + \Theta((\lg \lg n)^2)$

  Substitute $n = 2^m$:

$$T(2^m) = 3T(2^{m/2}) + 3T(2^{m/3}) + \Theta((\lg \lg(2^m))^2)$$

$$S(m) = 3S(m/2) + 3S(m/3) + \Theta(\lg^2 m)$$

  Apply Akra-Bazzi Theorem to $S(m)$: Find p such that $3\left(\frac{1}{2}\right)^p + 3\left(\frac{1}{3}\right)^p = 1$. For $p = 2$, $3(1/4) + 3(1/9) = 13/12 > 1 \implies p > 2$.

  Evaluate integral with $g(m) = \lg^2 m$:

$$\int_1^m \frac{\lg^2 u}{u^{p+1}} du$$

  Since $p > 2 \implies p + 1 > 3$, the integral converges to a constant $\Theta(1)$ as $m \to \infty$.

$$S(m) = \Theta\left(m^p(1 + \Theta(1))\right) = \Theta(m^p)$$

  Back-substitute $m = \lg n$:

$$T(n) = \Theta((\lg n)^p)$$

  (where $p \approx 2.11$ is the constant satisfying $3(1/2)^p + 3(1/3)^p = 1$)